

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 9: Consolidation and Digital Animations: Comparing Mitosis and Meiosis to Explain Growth

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	A	D	B	A	B

USE AFTER: PREDICT PHASE — Think-Pair-Share: 3-box prediction sketch of one cell dividing (before, during, after)

Q1. A cell in a growing child's leg bone has 46 chromosomes. It divides by mitosis to help the child grow. How many chromosomes will each new cell have?

- A. 23, because chromosomes are halved in every division
- B. 92, because the new cells add up the chromosomes
- C. 69, because each new cell gets half of the extra chromosomes
- D. 46, because each new cell gets a full copy of the set**

Correct answer: D. In mitosis the chromosomes are copied first and then shared equally, so each daughter cell has the same number as the parent cell (46). Halving happens only in meiosis.

USE AFTER: OBSERVE PHASE — Watching the mitosis animation and completing the stage-tracking sheet (prophase, metaphase, anaphase, telophase)

Q2. During mitosis, at which stage do the sister chromatids pull apart and move to opposite poles of the cell?

- A. Prophase
- B. Metaphase
- C. Telophase
- D. Anaphase**

Correct answer: D. In anaphase the spindle fibres pull the sister chromatids apart to opposite poles. In metaphase they only line up at the equator.

USE AFTER: OBSERVE PHASE — Jigsaw: home groups complete the mitosis vs meiosis comparison table

Q3. One diploid (2n) cell in the testes of a bull divides by meiosis. What is produced at the end?

- A. Four genetically different haploid cells**
- B. Two genetically different haploid cells
- C. Four genetically identical diploid cells
- D. Two genetically identical diploid cells

Correct answer: A. Meiosis has two divisions and gives four haploid (n) cells that are genetically different because of crossing over. Two identical diploid cells describes mitosis.

USE AFTER: EXPLAIN PHASE — Discussion Diamond and 4-sentence explanation: how one fertilised egg becomes billions of cells

Q4. A fertilised egg divides by mitosis. Every cell divides once in each round, and no cells die. How many cells will there be after 5 rounds of mitosis?

- A. 16 cells
- B. 25 cells
- C. 10 cells
- D. 32 cells**

Correct answer: D. The number of cells doubles in each round: 1, 2, 4, 8, 16, 32. After 5 rounds there are $2^5 = 32$ cells. Repeated rounds of mitosis can therefore give billions of cells.

USE AFTER: EXPLAIN PHASE — Discussion Diamond: teacher question 'What role does the food really play?'

Q5. A learner says, 'A child grows because the ugali, beans and sukuma wiki we eat turn directly into new cells.' Which statement best corrects this idea?

- A. Food supplies the chromosomes that are put into each new cell
- B. Food supplies energy and raw materials, but new cells come from existing cells dividing by mitosis**
- C. Food makes the existing cells bigger, and that is the only way the body grows
- D. Food is not needed because cells make everything they need by meiosis

Correct answer: B. Food gives the energy and raw materials needed to build new cells, but the cells themselves are made when existing cells divide by mitosis. Growth is not only cells getting bigger.

USE AFTER: DQB PHASE — Driving Question Board sorting and exit ticket, Prompt 2: which process helps a child grow, and why?

Q6. A girl falls and cuts her knee, and the cut heals after a few days. Which process makes the new skin cells that close the wound, and why?

- A. Mitosis, because it makes identical diploid cells for growth and repair**
- B. Meiosis, because it halves the chromosome number to make skin cells
- C. Mitosis, because it makes genetically different cells with crossing over
- D. Meiosis, because it makes four new cells from each cell

Correct answer: A. Mitosis produces two genetically identical diploid cells, which is how the body grows and repairs damaged tissue. Meiosis only makes gametes.

USE AFTER: MODEL BUILDING PHASE — Gallery Walk: groups present chromosome-behaviour models and give 'Two stars and a wish' feedback

Q7. A group makes a model of meiosis in a cow. A cow body cell has 60 chromosomes. How many chromosomes should their model show in each egg cell (gamete) at the end of meiosis?

- A. 120 chromosomes
- B. 30 chromosomes**
- C. 15 chromosomes
- D. 60 chromosomes

Correct answer: B. Meiosis halves the chromosome number from diploid ($2n = 60$) to haploid (n): $60 \div 2 = 30$. This keeps the number constant when the egg and sperm join at fertilisation.